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millimetre Wave Transmission (mWT); Conformance Test Specification for Wireless Transport Profile for Standard SDN Northbound Interfaces; Part 2: Test Suite Structure (TSS) and Test Purposes (TP)

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Foreword

This Group Specification (GS) has been produced by ETSI Industry Specification Group (ISG) millimetre Wave Transmission (mWT).

The present document is part 2 of a multi-part deliverable covering Conformance Test Specification for Wireless Transport Profile for Standard SDN Northbound Interfaces, as identified below:

Part 1: "Implementation Conformance Statement (ICS)";

Part 2: "Test Suite Structure (TSS) and Test Purposes (TP)";

Part 3: "Abstract Test Suite (ATS) and partial Implementation eXtra Information for Testing (PIXIT)".

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1 Scope

The present document provides the Test Suite Structure (TSS) and Test Purposes (TP) of a multi-part conformance test specification for the mWT Standard SDN northbound APIs for the exchange of messages, complying with the ETSI GS mWT 024 [1] specifications, in accordance with the relevant guidance given in ISO/IEC 9646-7 [i.4].

The ISO standards for the methodology of conformance testing (ISO/IEC 9646-1 [i.1] and ISO/IEC 9646-2 [i.2]) as well as the ETSI rules for conformance testing (ETSI ETS 300 406 [i.5]) are used as a basis for the test methodology.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found in the ETSI docbox.

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The following referenced documents are necessary for the application of the present document.

[1] ETSI GS mWT 024: "millimetre Wave Transmission (mWT); Definition of a Wireless Transport Profile for Standard SDN Northbound Interfaces".

[2] IETF RFC 8453: "Framework for Abstraction and Control of TE Networks (ACTN)".

[3] ETSI GS mWT 029-1: "millimetre Wave Transmission (mWT); Conformance Test Specification for Wireless Transport Profile for Standard SDN Northbound Interfaces; Part 1: Implementation Conformance Statement (ICS)".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

[i.1] ISO/IEC 9646-1 (1994): "Information technology — Open Systems Interconnection — Conformance testing methodology and framework — Part 1: General concepts".

[i.2] ISO/IEC 9646-2 (1994): "Information technology — Open Systems Interconnection — Conformance testing methodology and framework — Part 2: Abstract Test Suite specification".

[i.3] ISO/IEC 9646-6 (1994): "Information technology — Open Systems Interconnection — Conformance testing methodology and framework — Part 6: Protocol profile test specification".

[i.4] ISO/IEC 9646-7 (1995): "Information technology — Open Systems Interconnection — Conformance testing methodology and framework — Part 7: Implementation Conformance Statements".

[i.5] ETSI ETS 300 406 (1995): "Methods for testing and Specification (MTS); Protocol and profile conformance testing specifications; Standardization methodology".

[i.6] IETF RFC 8040: "RESTCONF Protocol".

[i.7] IETF RFC 8525: "YANG Library".

[i.8] IETF RFC 8345: "A YANG Data Model for Network Topologies".

[i.9] IETF RFC 8795: "YANG Data Model for Traffic Engineering (TE) Topologies".

[i.10] draft-ietf-ccamp-mw-topo-yang-02: "A YANG Data Model for Microwave Topology".

[i.11] IETF RFC 6991: "Common YANG Data Types".

[i.12] IETF RFC 8776: "Common YANG Data Types for Traffic Engineering".

[i.13] draft-ietf-ccamp-client-signal-yang-05: "A YANG Data Model for Transport Network Client Signals".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI GS mWT 024 [1], ISO/IEC 9646-6 [i.3] and ISO/IEC 9646-7 [i.4] apply.

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

API	Application Programming Interface
ATS	Abstract Test Suite
BO	Invalid Behavior
BV	Valid Behaviour
CBS	Committed Burst Size
CIR	Committed Information Rate
EBS	Excess Burst Size
EIR	Excess Information Rate
ETS	European Telecommunication Standard
ICS	Implementation eXtra Information for Testing
MDSC	Multi Domain Service Coordinator
MPI	MDSC-PNC Interface
MSDC	Massively Scalable Data Centers
MW	Microwave
mWT	millimetre Wave Transmission
N/A	Not Applicable
PICS	Partial Implementation Conformance Statement
PIXIT	Partial Implementation eXtra Information for Testing
PNC	Provisioning Network Controller
SDN	Software Defined Networking

TE Traffic Engineering URIUniform Resource Identifier
YANG Yet Another Next Generation

4 Test Suite Structure (TSS)

4.1 Structure for ETSI mWT Standard SDN northbound API tests

Table 1 shows the ETSI mWT Standard SDN northbound API Test Suite Structure (TSS) defined for conformance testing.

Table 1: TSS for mWT Standard SDN northbound API

Root	Group	Sub-group	Category
mWT	MDSC	MW_TOPO	Valid and Invalid
		ETH_TOPO	Valid
	PNC	MW_TOPO	Valid and Invalid
		ETH_TOPO	Valid

5 Test Purposes (TP)

5.1 Introduction

5.1.1 TP definition conventions

The TP definition is built according to ISO/IEC 9646-6 [i.3].

5.1.2 TP Identifier naming conventions

The identifier of the TP is built according to table 2.

Table 2: TP naming convention

Identifier	TP_<root>_<tgt>_<gr>_<sub-gr> <sn>_<x>	Sub-Group	Category
TP	<root> = root	mWT	mWT Conformance tests
	<tgt> = target	MDSC	Multi-Domain Service Coordinator
		PNC	Provisioning Network Controller
	<gr> = group	RESTCONF	RESTCONF support
		ABSTRACT_BB	Abstract Blackbox
		PARTIALLY_TRANS	Partially Transparent
	<gr> = sub-group	MW_TOPO	Microwave topology
		ETH_TOPO	Ethernet Topology
		ETH_SERVICE	Ethernet Service
	<sn> = test purpose sequential number		01 to 99
	<x> = category	BV	Valid Behaviour tests
		BO	Invalid Behaviour Tests

5.1.3 Rules for the behaviour description

The description of the TP is built according to ISO/IEC 9646-6 [i.3].

Being "idle" refers to the starting point of the initial device configuration. There are no pending actions, no instantiated buffers or variables, which could disturb the execution of a test.

5.1.4 Sources of TP definitions

All TPs have been specified according to ETSI GS mWT 024 [1] and IETF RFC 8453 [2].

5.1.5 Mnemonics for PICS reference

To avoid an update of all TPs when the PICS document is changed, table 3 introduces mnemonics name and the correspondence with the real PICS item number. The 'PICS item' as defined in ETSI GS mWT 024 [1] and IETF RFC 8453 [2] shall be used to determine the test applicability.

Table 3: Mnemonics for PICS reference

	Mnemonic	PICS item
1	IUT_SDN	[3] A.1/1
2	IUT_APP	[3] A.1/2
3	ABSTRACT_BB	[3] A.2/1
4	PARTIALLY_TRANS	[3] A.2/2
5	MW_TOPO	[3] A.3/1
6	ETH_TOPO	[3] A.3/2
7	SERVICE_TOPO	[3] A.3/3

5.2 Test Purposes for MSDC

N/A.

5.3 Test Purposes for PNC

5.3.1 RESTCONF support

TP Id	TP_mWT_PNC_RESTCONF_GET_BV_01
Test Objective	Verify that the IUT supports RESTCONF API request.
Reference	ETSI GS mWT 024 [1], clause 5.2.1 "RESTCONF Protocol" IETF RFC 8040 [i.6] "RESTCONF Protocol" IETF RFC 8525 [i.7] "YANG Library"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS
Initial Conditions	

with {
the IUT is idle
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a vRequestGet containing
uri indicating value "/restconf/data/ietf-yang-library:modules-state"
from the TEST_SYSTEM entity
} then {
the IUT sends a r200 containing
headers containing
content_type indicating value "application/yang-data+xml",
body containing
modules_state containing
name_ indicating value "ietf-yang-library:modules-state",
modules containing
non empty list
to the TEST_SYSTEM entity
}
}	

TP Id	TP_mWT_PNC_RESTCONF_GET_BV_02
Test Objective	Verify that the IUT supports the basic YANG data models.
Reference	ETSI GS mWT 024 [1], clause 5.2.1 "RESTCONF Protocol" ETSI GS mWT 024 [1], clause 5.2.2 "YANG Version" IETF RFC 8040 [i.6] "RESTCONF Protocol" IETF RFC 8525 [i.7] "YANG Library"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS

Initial Conditions

```
with {
the IUT is idle
               }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestGet containing
            uri indicating value "/restconf/data/ietf-yang-library:modules-state"
        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r200 containing
            headers containing
                content_type indicating value "application/yang-data+xml",
            body containing
                modules_state containing
                    name_ indicating value "ietf-yang-library:modules-state",
                    modules containing
                        entry containing
                            name_ indicating value "ietf-datastores",
                            revision indicating value "2018-02-14",
                            schema indicating value
"https://localhost:443/restconf/data/modules/ietf-datastores/2018-02-14",
                            namespace indicating value "urn:ietf:params:xml:ns:yang:ietf-
datastores",
                            conformance_type indicating value "import",
                        entry containing
                            name_ indicating value "ietf-inet-types",
                            revision indicating value "2013-07-15",
                            schema indicating value
"https://localhost:443/restconf/data/modules/ietf-inet-types/2013-07-15",
                            namespace indicating value "urn:ietf:params:xml:ns:yang:ietf-inet-
types",
                            conformance_type indicating value "import",
                        entry containing
                            name_ indicating value "ietf-interfaces",
                            revision indicating value "2018-02-20",
                            schema indicating value
"https://localhost:443/restconf/data/modules/ietf-interfaces/2018-02-20",
                            namespace indicating value "urn:ietf:params:xml:ns:yang:ietf-
interfaces",
                            conformance_type indicating value "import",
                        entry containing
                            name_ indicating value "ietf-ip",
                            revision indicating value "2018-02-22",
                            schema indicating value
"https://localhost:443/restconf/data/modules/ietf-ip/2018-02-22",
                            namespace indicating value "urn:ietf:params:xml:ns:yang:ietf-ip",
                            conformance_type indicating value "import",
                        entry containing
                            name_ indicating value "ietf-network-instance",
                            revision indicating value "2019-01-21",
                            schema indicating value
"https://localhost:443/restconf/data/modules/ietf-network-instance/2019-01-21",
                            namespace indicating value "urn:ietf:params:xml:ns:yang:ietf-network-
instance",
                            conformance_type indicating value "import",
                        entry containing
                            name_ indicating value "ietf-restconf",
                            revision indicating value "2017-01-26",
                            schema indicating value
"https://localhost:443/restconf/data/modules/ietf-restconf/2017-01-26",
                            namespace indicating value "urn:ietf:params:xml:ns:yang:ietf-
restconf",
                            conformance_type indicating value "implement",
                        entry containing
                            name_ indicating value "ietf-yang-types",
                            revision indicating value "2013-07-15",
```

```
                            schema indicating value
"https://localhost:443/restconf/data/modules/ietf-restconf/2017-01-26",
                            namespace indicating value "urn:ietf:params:xml:ns:yang:ietf-
restconf",
                            conformance_type indicating value "import",
                        entry containing
                            name_ indicating value "ietf-yang-library",
                            revision indicating value "2019-01-04",
                            schema indicating value
"https://localhost:443/restconf/data/modules/ietf-yang-types/2013-07-15",
                            namespace indicating value "urn:ietf:params:xml:ns:yang:ietf-yang-
types",
                            conformance_type indicating value "implement"
        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_RESTCONF_GET_BV_03
Test Objective	Verify that the IUT provides the list of supported ETSI GS mWT 024 [1] YANG data models.
Reference	ETSI GS mWT 024 [1], clause 5.2.2.2 "Common YANG data models" ETSI GS mWT 024 [1], clause 5.2.2.3 "YANG Data Models at the Microwave MPI" ETSI GS mWT 024 [1], clause 5.2.2.4 "Hierarchical relationship among YANG data models"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS
Initial Conditions	

```
with {
the IUT is idle
                }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestGet containing
            uri indicating value "/restconf/data/ietf-yang-library:modules-state"
        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r200 containing
            body containing
                modules_state containing
                    modules containing
                        entry containing
                            name_ indicating value "ietf-microwave-topology",
                            revision indicating value "2021-10-12",
                            schema indicating value "draft-ietf-ccamp-mw-topo-yang-02",
                            namespace indicating value "",
                            conformance_type indicating value "implement",
                        entry containing
                            name_ indicating value "ietf-eth-te-topology",
                            revision indicating value "2019-11-18",
                            schema indicating value "draft-ietf-ccamp-eth-client-te-topo-yang-01",
                            namespace indicating value "",
                            conformance_type indicating value "implement",
                        entry containing
                            name_ indicating value "idraft-ietf-ccamp-client-signal-yang-06",
                            revision indicating value "",
                            schema indicating value "draft-ietf-ccamp-client-signal-yang-06",
                            namespace indicating value "",
                            conformance_type indicating value ""
        to the TEST_SYSTEM entity
    }
}
```

5.3.2 Black-box abstraction level

TP Id	TP_mWT_PNC_BB_MW_TOPO_GET_BV_01
Test Objective	Verify that the IUT does not provide any MW network topology.
Reference	ETSI GS mWT 024 [1], clause 4.3.2 ETSI GS mWT 024 [1], clause A.2.3 IETF RFC 8345 [i.8] "A YANG Data Model for Network Topologies" IETF RFC 8795 [i.9] "YANG Data Model for Traffic Engineering (TE) Topologies"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_BB
Initial Conditions	

```
with {
the IUT is idle
                }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestGet containing
            uri indicating value "/restconf/data/ietf-network:networks"
        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r200 containing
        headers containing
            content_type indicating value "application/yang-data+xml",
        body containing
            networks containing
                networksList containing
                    name_ indicating value any_value,
                    networks containing
                        network_id indicating value any_value,
                        network_types containing
                            not te_topology
                            ,
                        node_id indicating value any_value,
                        termination_points indicating value any_value,
                        links indicating value any_value

        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_BB_MW_TOPO_GET_BV_01
Test Objective	Verify that the IUT provides a network list containing only one node.
Reference	ETSI GS mWT 024 [1], clause 4.3.2 ETSI GS mWT 024 [1], clause A.2.3 IETF RFC 8345 [i.8] "A YANG Data Model for Network Topologies" IETF RFC 8795 [i.9] "YANG Data Model for Traffic Engineering (TE) Topologies"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_BB
Initial Conditions	

```
with {
the IUT is idle
                }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestGet containing
            uri indicating value "/restconf/data/ietf-network:networks"
        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r200 containing
        headers containing
            content_type indicating value "application/yang-data+xml",
        body containing
            networks containing
                networksList containing
                     one network node

        to the TEST_SYSTEM entity
    }
}
```

5.3.3 Partially Transparent abstraction level

5.3.3.1 Microwave topology inventory

TP Id	TP_mWT_PNC_PT_MW_TOPO_GET_BV_01
Test Objective	Verify that the IUT provides the MW network topology for all networks with mandatory ETSI GS mWT 024 [1] data elements subset.
Reference	ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause A.2.2 IETF RFC 8345 [i.8] "A YANG Data Model for Network Topologies" IETF RFC 8795 [i.9] "YANG Data Model for Traffic Engineering (TE) Topologies" draft-ietf-ccamp-mw-topo-yang-02 [i.10] "A YANG Data Model for Microwave Topology"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT and MW_TOPO
Initial Conditions	

```
with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestGet containing
            uri indicating value "/restconf/data/ietf-network:networks/topologyId/mw-native-
topology",
            headers containing
                content_type indicating value "application/yang-data+xml"
        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r200 containing
        headers containing
            content_type indicating value "application/yang-data+xml",
        body containing
            networks containing
                networksList containing
                    name_ indicating value any_value,
                    networks containing
                        network_id indicating value any_value,
                        network_types containing
                            te_topology containing
                                mw_topology containing
                                    mw_tp_choice indicating value any_value
                        ,
                        node_id indicating value any_value,
                        termination_points indicating value any_value,
                        links indicating value any_value

        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_PT_MW_TOPO_GET_BV_02
Test Objective	Verify that the IUT provides the MW network topology for a specific network with mandatory ETSI GS mWT 024 [1] data elements subset.
Reference	ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause A.2.2.1 IETF RFC 8345 [i.8] "A YANG Data Model for Network Topologies" IETF RFC 8795 [i.9] "YANG Data Model for Traffic Engineering (TE) Topologies"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT and MW_TOPO
Initial Conditions	

```
with {
the IUT is idle and
the IUT having discovered a network topology NETWORK_TOPO
                    }
```

Expected Behaviour

ensure that {
  when {
  the IUT receives a vRequestGet containing
  uri indicating value "/restconf/data/ietf-microwave-topology/ietf-network:networks/network={NETWORK_ID/topologyId/mw-native-topology",
  headers containing
  content_type indicating value "application/yang-data+xml"
  from the TEST_SYSTEM entity
  } then {
  the IUT sends a r200 containing
  headers containing
  content_type indicating value "application/yang-data+xml",
  body containing
  networks containing
  network_id indicating value NETWORK_ID,
  network_types containing
  te_topology containing
  mw_topology containing
  mw_tp_choice indicating value any_value
  ,
  node_id indicating value any_value,
  termination_points indicating value any_value,
  links indicating value any_value
 	
  to the TEST_SYSTEM entity
  }
}

TP Id	TP_mWT_PNC_PT_MW_TOPO_GET_BO_01
Test Objective	Verify that the IUT replies with an error code when the request is malformed.
Reference	ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause 5.3.2 "Exception Handling" ETSI GS mWT 024 [1], clause A.2.2 IETF RFC 8345 [i.8] "A YANG Data Model for Network Topologies" IETF RFC 8795 [i.9] "YANG Data Model for Traffic Engineering (TE) Topologies"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT and MW_TOPO
Initial Conditions	

with {
the IUT is idle and
the IUT having discovered a network topology NETWORK_TOPO
}	

Expected Behaviour	

ensure that {
  when {
  the IUT receives a vRequestGet containing
  uri indicating value "/restconf/data/networks" // Malformed URI
  from the TEST_SYSTEM entity
  } then {
  the IUT sends a r400
  to the TEST_SYSTEM entity
  }
}	

TP Id	TP_mWT_PNC_PT_MW_TOPO_GET_BO_02
Test Objective	Verify that the IUT replies with an error code when the network identifier is unknown.
Reference	ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause 5.3.2 "Exception Handling" ETSI GS mWT 024 [1], clause A.2.2 IETF RFC 8345 [i.8] "A YANG Data Model for Network Topologies" IETF RFC 8795 [i.9] "YANG Data Model for Traffic Engineering (TE) Topologies"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT and MW_TOPO

Initial Conditions

with {
the IUT is idle and
the IUT having discovered a network topology NETWORK_TOPO
}
Expected Behaviour
ensure that {
when {
the IUT receives a vRequestGet containing
uri indicating value "/restconf/data/ietf-network:networks/network={UNKNOWN_NETWORK_ID/topologyId/mw-native-topology"
from the TEST_SYSTEM entity
} then {
the IUT sends a r404
to the TEST_SYSTEM entity
}
}

TP Id	TP_mWT_PNC_PT_MW_NODE_GET_BV_01
Test Objective	Verify that the IUT provides the MW network topology for all nodes with mandatory ETSI GS mWT 024 [1] data elements subset.
Reference	ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause A.2.2.1 IETF RFC 8345 [i.8] "A YANG Data Model for Network Topologies" IETF RFC 8795 [i.9] "YANG Data Model for Traffic Engineering (TE) Topologies" draft-ietf-ccamp-mw-topo-yang-02 [i.10] "A YANG Data Model for Microwave Topology"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT and MW_TOPO

Initial Conditions

```
with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestGet containing
            uri indicating value "/restconf/data/ietf-microwave-topology/ietf-network:networks/network={NETWORK_ID/node/topologyId/mw-native-topology"
        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r200 containing
        headers containing
            content_type indicating value "application/yang-data+xml",
        body containing
            networks containing
                networksList containing
                    name_ indicating value any_value,
                    networks containing
                        network_id indicating value any_value,
                        network_types containing
                            te_topology containing
                                mw_native_topology indicating value any_value
                            ,
                        node_id indicating value any_value,
                        termination_points indicating value any_value,
                        links indicating value any_value

        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_PT_MW_NODE_GET_BV_02
Test Objective	Verify that the IUT provides the MW network topology for a specific node with mandatory ETSI GS mWT 024 [1] data elements subset.
Reference	ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause A.2.2.1 IETF RFC 8345 [i.8] "A YANG Data Model for Network Topologies" IETF RFC 8795 [i.9] "YANG Data Model for Traffic Engineering (TE) Topologies" draft-ietf-ccamp-mw-topo-yang-02 [i.10] "A YANG Data Model for Microwave Topology"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT and MW_TOPO
Initial Conditions	

```
with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestGet containing
            uri indicating value "/restconf/data/ietf-microwave-topology/ietf-
network:networks/network={NETWORK_ID/node={NODE_ID}/topologyId/mw-native-topology"
        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r200 containing
        headers containing
            content_type indicating value "application/yang-data+xml",
        body containing
            networks containing
                networksList containing
                    name_ indicating value any_value,
                    networks containing
                        network_id indicating value NETWORK_ID,
                        network_types containing
                            te_topology containing
                                mw_native_topology indicating value any_value
                            ,
                        node_id indicating value NODE_ID,
                        termination_points indicating value any_value,
                        links indicating value any_value

        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_PT_MW_NODE_GET_BO_01
Test Objective	Verify that the IUT replies with an error code when the node identifier is unknown.
Reference	ETSI GS mWT 024 [1] clause 4.3.3 ETSI GS mWT 024 [1] clause 5.3.2 "Exception Handling" ETSI GS mWT 024 [1] clause A.2.2 IETF RFC 8345 [i.8] "A YANG Data Model for Network Topologies" IETF RFC 8795 [i.9] "YANG Data Model for Traffic Engineering (TE) Topologies" draft-ietf-ccamp-mw-topo-yang-02 [i.10] "A YANG Data Model for Microwave Topology"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT and MW_TOPO
Initial Conditions	

```
with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestGet containing
            uri indicating value "/restconf/data/ietf-microwave-topology/ietf-
network:networks/network={NETWORK_ID/node={UNKNOWN_NODE_ID}/topologyId/mw-native-topology"
        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r404
        to the TEST_SYSTEM entity
    }
}
```

5.3.3.2 Ethernet topology inventory

TP Id	TP_mWT_PNC_PT_ETH_TOPO_GET_BV_01
Test Objective	Verify that the IUT provides the Ethernet network topology for all networks with mandatory ETSI GS mWT 024 [1] data elements subset.
Reference	ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause A.2.2 IETF RFC 8345 [i.8] "A YANG Data Model for Network Topologies" IETF RFC 8795 [i.9] "YANG Data Model for Traffic Engineering (TE) Topologies" draft-ietf-ccamp-mw-topo-yang-02 [i.10] "A YANG Data Model for Microwave Topology"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT and ETH_TOPO
Initial Conditions	

```
with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestGet containing
            uri indicating value "/restconf/data/ietf-network:networks",
            headers containing
                content_type indicating value "application/yang-data+xml"
        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r200 containing
        headers containing
            content_type indicating value "application/yang-data+xml",
        body containing
            networks containing
                networksList containing
                    name_ indicating value any_value,
                    networks containing
                        network_id indicating value any_value,
                        network_types containing
                            te_topology containing
                                mw_topology containing
                                    mw_tp_choice indicating value any_value
                        ,
                        node_id indicating value any_value,
                        termination_points indicating value any_value,
                        links indicating value any_value

        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_PT_ETH_TOPO_GET_BV_02
Test Objective	Verify that the IUT provides the Ethernet network topology for a specific network with mandatory ETSI GS mWT 024 [1] data elements subset.
Reference	ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause A.1.2 IETF RFC 8345 [i.8] "A YANG Data Model for Network Topologies" IETF RFC 8795 [i.9] "YANG Data Model for Traffic Engineering (TE) Topologies"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT and ETH_TOPO
Initial Conditions	

```
with {
the IUT is idle and
the IUT having discovered a network topology NETWORK_TOPO
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestGet containing
            uri indicating value "/restconf/data/ietf-network:networks/network={NETWORK_ID",
            headers containing
                content_type indicating value "application/yang-data+xml"
        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r200 containing
        headers containing
            content_type indicating value "application/yang-data+xml",
        body containing
            networks containing
                network_id indicating value NETWORK_ID,
                network_types containing
                    te_topology indicating value any_value,
                node_id indicating value any_value,
                termination_points indicating value any_value,
                links indicating value any_value

        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_PT_ETH_TOPO_GET_BO_01
Test Objective	Verify that the IUT replies with an error code when the request is malformed.
Reference	ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause 5.3.2 "Exception Handling" ETSI GS mWT 024 [1], clause A.1.2 IETF RFC 8345 [i.8] "A YANG Data Model for Network Topologies" IETF RFC 8795 [i.9] "YANG Data Model for Traffic Engineering (TE) Topologies"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT and ETH_TOPO

Initial Conditions

```
with {
the IUT is idle and
the IUT having discovered a network topology NETWORK_TOPO
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestGet containing
            uri indicating value "/restconf/data/networks/NETWORK_ID" // Malformed URI
        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r400
        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_PT_ETH_TOPO_GET_BO_02
Test Objective	Verify that the IUT replies with an error code when the network identifier is unknown.
Reference	ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause 5.3.2 "Exception Handling" ETSI GS mWT 024 [1], clause A.1.2 IETF RFC 8345 [i.8] "A YANG Data Model for Network Topologies" IETF RFC 8795 [i.9] "YANG Data Model for Traffic Engineering (TE) Topologies"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT and ETH_TOPO

Initial Conditions

```
with {
the IUT is idle and
the IUT having discovered a network topology NETWORK_TOPO
                    }
```

Expected Behaviour

ensure that {
when {
the IUT receives a vRequestGet containing
uri indicating value "/restconf/data/ietf-network:networks/network={UNKNOWN_NETWORK_ID"
from the TEST_SYSTEM entity
} then {
the IUT sends a r404
to the TEST_SYSTEM entity
}
}

TP Id	TP_mWT_PNC_PT_ETH_NODE_GET_BV_01
Test Objective	Verify that the IUT provides the Ethernet network topology for all nodes with mandatory ETSI GS mWT 024 [1] data elements subset.
Reference	ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause A.2.2 IETF RFC 8345 [i.8] "A YANG Data Model for Network Topologies" IETF RFC 8795 [i.9] "YANG Data Model for Traffic Engineering (TE) Topologies" draft-ietf-ccamp-mw-topo-yang-02 [i.10] "A YANG Data Model for Microwave Topology"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT and ETH_TOPO

Initial Conditions

with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies
}

Expected Behaviour

ensure that {
when {
the IUT receives a vRequestGet containing
uri indicating value "/restconf/data/ietf-network:networks/network={NETWORK_ID/node"
from the TEST_SYSTEM entity
} then {
the IUT sends a r200 containing
headers containing
content_type indicating value "application/yang-data+xml",
body containing
networks containing
networksList containing
name_ indicating value any_value,
networks containing
network_id indicating value any_value,
network_types containing
te_topology containing
mw_topology containing
mw_tp_choice indicating value any_value
,
node_id indicating value any_value,
termination_points indicating value any_value,
links indicating value any_value
 	
to the TEST_SYSTEM entity
}
}

TP Id	TP_mWT_PNC_PT_ETH_NODE_GET_BV_02
Test Objective	Verify that the IUT provides the Ethernet network topology for specific node with mandatory ETSI GS mWT 024 [1] data elements subset.
Reference	ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause A.2.2 IETF RFC 8345 [i.8] "A YANG Data Model for Network Topologies" IETF RFC 8795 [i.9] "YANG Data Model for Traffic Engineering (TE) Topologies" draft-ietf-ccamp-mw-topo-yang-02 [i.10] "A YANG Data Model for Microwave Topology"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT and ETH_TOPO

Initial Conditions

```
with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestGet containing
            uri indicating value "/restconf/data/ietf-
network:networks/network={NETWORK_ID/node={NODE_ID"
        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r200 containing
        headers containing
            content_type indicating value "application/yang-data+xml",
        body containing
            networks containing
                networksList containing
                    name_ indicating value any_value,
                    networks containing
                        network_id indicating value any_value,
                        network_types containing
                            te_topology containing
                                mw_topology containing
                                    mw_tp_choice indicating value any_value
                        ,
                        node_id indicating value any_value,
                        termination_points indicating value any_value,
                        links indicating value any_value

        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_PT_ETH_NODE_GET_BO_01
Test Objective	Verify that the IUT replies with an error code when the node identifier is unknown.
Reference	ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause 5.3.2 "Exception Handling" ETSI GS mWT 024 [1], clause A.2.2 IETF RFC 8345 [i.8] "A YANG Data Model for Network Topologies" IETF RFC 8795 [i.9] "YANG Data Model for Traffic Engineering (TE) Topologies" draft-ietf-ccamp-mw-topo-yang-02 [i.10] "A YANG Data Model for Microwave Topology"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT and ETH_TOPO

Initial Conditions

```
with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestGet containing
            uri indicating value "/restconf/data/ietf-
network:networks/network={NETWORK_ID/node={UNKNOWN_NODE_ID"
        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r404
        to the TEST_SYSTEM entity
    }
}
```

5.3.3.3 Ethernet services

TP Id	TP_mWT_PNC_PT_ETH_SVC_GET_BV_01
Test Objective	Verify that the IUT provides all the Bandwidth profiles with mandatory ETSI GS mWT 024 [1] data elements subset.
Reference	ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause A.2.2 IETF RFC 8345 [i.8] "A YANG Data Model for Network Topologies" IETF RFC 8795 [i.9] "YANG Data Model for Traffic Engineering (TE) Topologies" draft-ietf-ccamp-client-signal-yang-05 [i.13] "A YANG Data Model for Transport Network Client Signals"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT
Initial Conditions	

```
with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies and
the IUT having a Bandwidth profile created and
the IUT having a service profile created
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestGet containing
            uri indicating value "/restconf/data/ietf-eth-tran-service:etht-svc/globals"
        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r200 containing
        headers containing
            content_type indicating value "application/yang-data+xml",
        body containing
            eth_svc containing
                globals containing
                    band_width_profiles containing
                        band_width_profile indicating value BAND_WIDTH_PROFILE
                ,
                ethtSvcInstanceList containing
                    ethtSvcInstance containing
                        name_ indicating value any_value,
                        title indicating value any_value,
                        desc indicating value any_value,
                        customer indicating value any_value

        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_PT_ETH_SVC_GET_BO_01
Test Objective	Verify that the IUT replies with an error code when the request is malformed.
Reference	ETSI GS mWT 024 [1], clause 4.2.2.1 "Ethernet Ethernet Services" ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause 5.3.2 "Exception Handling" ETSI GS mWT 024 [1], clause A.2.4 IETF RFC 6991 [i.11] "Common YANG Data Types" IETF RFC 8776 [i.12] "Common YANG Data Types for Traffic Engineering" draft-ietf-ccamp-client-signal-yang-05 [i.13] "A YANG Data Model for Transport Network Client Signals"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT
Initial Conditions	

```
with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestGet containing
            uri indicating value "/restconf/data/tran-service:etht-svc/globals" // Malformed URI
        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r400
        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_PT_ETH_SVC_POST_BV_01
Test Objective	Verify that the IUT creates a new Bandwidth profile with mandatory ETSI GS mWT 024 [1] data elements subset.
Reference	ETSI GS mWT 024 [1], clause 4.2.2.1 "Ethernet Ethernet Services" ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause A.2.4 IETF RFC 6991 [i.11] "Common YANG Data Types" IETF RFC 8776 [i.12] "Common YANG Data Types for Traffic Engineering" draft-ietf-ccamp-client-signal-yang-05 [i.13] "A YANG Data Model for Transport Network Client Signals"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT and not BWP_SVS_IN_ONE_REQUEST

Initial Conditions

```
with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies and
the IUT not having a Bandwidth profile created
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestPost containing
            uri indicating value "/restconf/data/ietf-eth-tran-service:etht-svc/globals",
            body containing
                eth_svc containing
                    ethtSvcInstanceList containing
                        ethtSvcInstance containing
                            name_ indicating value ETHT_SVC_NAME,
                            title indicating value any_value,
                            desc indicating value any_value,
                            customer indicating value any_value

        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r201 containing
        headers containing
            content_type indicating value "application/yang-data+xml",
        body containing
            eth_svc containing
                ethtSvcInstanceList containing
                    ethtSvcInstance containing
                        name_ indicating value ETHT_SVC_NAME,
                        title indicating value any_value,
                        desc indicating value any_value,
                        customer indicating value any_value

        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_PT_ETH_SVC_POST_BV_02_01
Test Objective	Verify that the IUT creates a new ethernet service with mandatory ETSI GS mWT 024 [1] data elements subset.
Reference	ETSI GS mWT 024 [1], clause 4.2.2.1 "Ethernet Ethernet Services" ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause A.2.4 IETF RFC 6991 [i.11] "Common YANG Data Types" IETF RFC 8776 [i.12] "Common YANG Data Types for Traffic Engineering" draft-ietf-ccamp-client-signal-yang-05 [i.13] "A YANG Data Model for Transport Network Client Signals"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT and not BWP_SVS_IN_ONE_REQUEST

Initial Conditions

```
with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies and
the IUT having a Bandwidth profile created and
the IUT not having a service profile created
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestPost containing
            uri indicating value "/restconf/data/ietf-eth-tran-service:etht-svc",
            headers containing
                content_type indicating value "application/yang-data+xml",
            body containing
                eth_svc containing
                    ethtSvcInstanceList containing
                        ethtSvcInstance containing
                            name_ indicating value ETHT_SVC_NAME,
                            title indicating value any_value,
                            desc indicating value any_value,
                            customer indicating value any_value

        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r201 containing
        headers containing
            content_type indicating value "application/yang-data+xml",
        body containing
            eth_svc containing
                ethtSvcInstanceList containing
                    ethtSvcInstance containing
                        name_ indicating value ETHT_SVC_NAME,
                        title indicating value any_value,
                        desc indicating value any_value,
                        customer indicating value any_value
                    ,
                    ethtSvcInstance containing
                        name_ indicating value ETHT_SVC_NAME,
                        title indicating value ETH_SVC_TITLE,
                        desc indicating value ETH_SVC_DESC,
                        customer indicating value ETH_SVC_CUSTOMER

        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_PT_ETH_SVC_POST_BV_02_02
Test Objective	Verify that the IUT creates successfully a new Bandwidth profile.
Reference	ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause A.2.2 IETF RFC 8345 [i.8] "A YANG Data Model for Network Topologies" IETF RFC 8795 [i.9] "YANG Data Model for Traffic Engineering (TE) Topologies" draft-ietf-ccamp-client-signal-yang-05 [i.13] "A YANG Data Model for Transport Network Client Signals"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT and not BWP_SVS_IN_ONE_REQUEST

Initial Conditions

```
with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies and
the IUT having a Bandwidth profile created and
the IUT not having a Bandwidth profile containing
    bandwidthProfileName indicating value SDN_BANDWIDTH_PROFILE_NAME and
the IUT having a service profile created
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestPost containing
            uri indicating value "/restconf/data/ietf-eth-tran-service:etht-svc/globals",
            headers containing
                content_type indicating value "application/yang-data+xml",
            body containing
                bandwidthProfiles containing
                    bandwidthProfileName indicating value SDN_BANDWIDTH_PROFILE_NAME,
                    bandwidthProfileType indicating value SDN_SERVICE_TYPE,
                    bandwidthProfileCir indicating value CIR,
                    bandwidthProfileEir indicating value EIR,
                    bandwidthProfileColorAware indicating value COLOR_AWARE,
                    bandwidthCouplingFlag indicating value true

        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r201 containing
        headers containing
            content_type indicating value "application/yang-data+xml",
        body containing
            bandwidthProfiles containing
                bandwidthProfileName indicating value SDN_BANDWIDTH_PROFILE_NAME,
                bandwidthProfileType indicating value SDN_SERVICE_TYPE,
                bandwidthProfileCir indicating value CIR,
                bandwidthProfileEir indicating value EIR,
                bandwidthProfileColorAware indicating value COLOR_AWARE,
                bandwidthCouplingFlag indicating value true

        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_PT_ETH_SVC_POST_BV_03
Test Objective	Verify that the IUT creates a new Bandwidth profile and a new ethernet service in one POST with mandatory ETSI GS mWT 024 [1] data elements subset.
Reference	ETSI GS mWT 024 [1], clause 4.2.2.1 "Ethernet Services" ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause A.2.4 IETF RFC 6991 [i.11] "Common YANG Data Types" IETF RFC 8776 [i.12] "Common YANG Data Types for Traffic Engineering" draft-ietf-ccamp-client-signal-yang-05 [i.13] "A YANG Data Model for Transport Network Client Signals"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT and BWP_SVS_IN_ONE_REQUEST

Initial Conditions

```
with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies and
the IUT not having a Bandwidth profile created
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestPost containing
            uri indicating value "/restconf/data/ietf-eth-tran-service:etht-svc/globals",
            headers containing
                content_type indicating value "application/yang-data+xml",
            body containing
                eth_svc containing
                    ethtSvcInstanceList containing
                        ethtSvcInstance containing
                            name_ indicating value ETHT_SVC_NAME,
```

```
                            title indicating value ETH_SVC_TITLE,
                            desc indicating value ETH_SVC_DESC,
                            customer indicating value ETH_SVC_CUSTOMER,
                            ingress_egress_bandwidth_profile containing
                                bandwidth_profile_type indicating value BANDWIDTH_PROFILE_TYPE,
                                cir indicating value CIR,
                                eir indicating value EIR

        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r201 containing
        headers containing
            content_type indicating value "application/yang-data+xml",
        body containing
            eth_svc containing
                ethtSvcInstanceList containing
                    ethtSvcInstance containing
                        name_ indicating value ETHT_SVC_NAME,
                        title indicating value ETH_SVC_TITLE,
                        desc indicating value ETH_SVC_DESC,
                        customer indicating value ETH_SVC_CUSTOMER

        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_PT_ETH_SVC_POST_BO_01
Test Objective	Verify that the IUT replies with an error when the request is malformed.
Reference	ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause 5.3.2 "Exception Handling" ETSI GS mWT 024 [1], clause A.2.4 IETF RFC 6991 [i.11] "Common YANG Data Types" IETF RFC 8776 [i.12] "Common YANG Data Types for Traffic Engineering" draft-ietf-ccamp-client-signal-yang-05 [i.13] "A YANG Data Model for Transport Network Client Signals"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT and not BWP_SVS_IN_ONE_REQUEST

Initial Conditions

```
with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestPost containing
            uri indicating value "/restconf/data/etht-svc/globals", // Malformed URI
            body containing
                eth_svc containing
                    ethtSvcInstanceList containing
                        ethtSvcInstance containing
                            name_ indicating value ETHT_SVC_NAME,
                            title indicating value ETH_SVC_TITLE,
                            desc indicating value ETH_SVC_DESC,
                            customer indicating value ETH_SVC_CUSTOMER

        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r400
        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_PT_ETH_SVC_PUT_BV_01
Test Objective	Verify that the IUT updates an existing Bandwidth profile with mandatory ETSI GS mWT 024 [1] data elements subset.
Reference	ETSI GS mWT 024 [1], clause 4.2.2.1 "Ethernet Services" ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause A.2.4 IETF RFC 6991 [i.11] "Common YANG Data Types" IETF RFC 8776 [i.12] "Common YANG Data Types for Traffic Engineering" draft-ietf-ccamp-client-signal-yang-05 [i.13] "A YANG Data Model for Transport Network Client Signals"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT

Initial Conditions

```
with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies and
the IUT having a Bandwidth profile created
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestPut containing
            uri indicating value "/restconf/data/ietf-eth-tran-service:etht-svc/globals",
            headers containing
                content_type indicating value "application/yang-data+xml",
            body containing
                eth_svc containing
                    globals containing
                        namedBandwidthProfile containing
                            bandwidth_profile_name indicating value BANDWIDTH_PROFILE_NAME,
                            CIR indicating value CIR_VALUE,
                            CBS indicating value CBS_VALUE,
                            EIR indicating value EIR_VALUE,
                            EBS indicating value EBS_VALUE,
                            color_aware indicating value COLOR_AWARE,
                            coupling_flag indicating value COUPLING_FLAG

        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r201 containing
        headers containing
            content_type indicating value "application/yang-data+xml",
        body containing
            eth_svc containing
                globals containing
                    namedBandwidthProfile containing
                        bandwidth_profile_name indicating value BANDWIDTH_PROFILE_NAME,
                        CIR indicating value CIR_VALUE,
                        CBS indicating value CBS_VALUE,
                        EIR indicating value EIR_VALUE,
                        EBS indicating value EBS_VALUE,
                        color_aware indicating value COLOR_AWARE,
                        coupling_flag indicating value COUPLING_FLAG

        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_PT_ETH_SVC_PUT_BV_02
Test Objective	Verify that the IUT updates an existing ethernet service.
Reference	ETSI GS mWT 024 [1], clause 4.2.2.1 "Ethernet Services" ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause A.2.4 IETF RFC 6991 [i.11] "Common YANG Data Types" IETF RFC 8776 [i.12] "Common YANG Data Types for Traffic Engineering" draft-ietf-ccamp-client-signal-yang-05 [i.13] "A YANG Data Model for Transport Network Client Signals"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT

Initial Conditions

```
with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies and
the IUT having a Bandwidth profile created
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestPut containing
            uri indicating value "/restconf/data/ietf-eth-tran-service:etht-svc/globals",
            headers containing
                content_type indicating value "application/yang-data+xml",
            body containing
                eth_svc containing
                    ethtSvcInstanceList containing
                        ethtSvcInstance containing
                            name_ indicating value ETHT_SVC_NAME,
                            title indicating value ETH_SVC_TITLE,
                            desc indicating value ETH_SVC_DESC,
                            customer indicating value ETH_SVC_CUSTOMER

        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r201 containing
        headers containing
            content_type indicating value "application/yang-data+xml",
        body containing
            eth_svc containing
                ethtSvcInstanceList containing
                    ethtSvcInstance containing
                        name_ indicating value ETHT_SVC_NAME,
                        title indicating value ETH_SVC_TITLE,
                        desc indicating value ETH_SVC_DESC,
                        customer indicating value ETH_SVC_CUSTOMER

        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_PT_ETH_SVC_PUT_BO_01
Test Objective	Verify that the IUT replies with an error code when updating an unknown Bandwidth profile with mandatory ETSI GS mWT 024 [1] data elements subset.
Reference	ETSI GS mWT 024 [1], clause 5.3.2 "Exception Handling" ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause A.2.4 IETF RFC 6991 [i.11] "Common YANG Data Types" IETF RFC 8776 [i.12] "Common YANG Data Types for Traffic Engineering" draft-ietf-ccamp-client-signal-yang-05 [i.13] "A YANG Data Model for Transport Network Client Signals"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT

Initial Conditions

```
with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestPut containing
            uri indicating value "/restconf/data/ietf-eth-tran-service:etht-svc/globals",
            body containing
                eth_svc containing
                    ethtSvcInstanceList containing
                        ethtSvcInstance containing
                            name_ indicating value UNKNOWN_ETHT_SVC_NAME,
                            title indicating value ETH_SVC_TITLE,
                            desc indicating value ETH_SVC_DESC,
                            customer indicating value ETH_SVC_CUSTOMER

        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r404
```

```
        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_PT_ETH_SVC_DELETE_BV_01
Test Objective	Verify that the IUT deletes an existing ethernet service.
Reference	ETSI GS mWT 024 [1], clause 4.2.2.1 "Ethernet Services" ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause A.2.4 IETF RFC 6991 [i.11] "Common YANG Data Types" IETF RFC 8776 [i.12] "Common YANG Data Types for Traffic Engineering" draft-ietf-ccamp-client-signal-yang-05 [i.13] "A YANG Data Model for Transport Network Client Signals"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT

Initial Conditions

```
with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies and
the IUT having a Bandwidth profile created and
the IUT having a service profile created
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestDelete containing
            uri indicating value "/restconf/data/ietf-eth-tran-service:etht-svc/etht-svc-
instances=ETHT_SVC_NAME",
            headers containing
                content_type indicating value "application/yang-data+xml"
        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r204
        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_PT_ETH_SVC_DELETE_BV_02
Test Objective	Verify that the IUT deletes an existing bandwidth profile.
Reference	ETSI GS mWT 024 [1], clause 4.2.2.1 "Ethernet Services" ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause A.2.4 IETF RFC 6991 [i.11] "Common YANG Data Types" IETF RFC 8776 [i.12] "Common YANG Data Types for Traffic Engineering" draft-ietf-ccamp-client-signal-yang-05 [i.13] "A YANG Data Model for Transport Network Client Signals"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT

Initial Conditions

```
with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies and
the IUT having a Bandwidth profile created
                    }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a vRequestDelete containing
            uri indicating value "/restconf/data/ietf-eth-tran-service:etht-
svc/globals/bandwitdh_profile_name=BANDWIDTH_PROFILE_NAME",
            headers containing
                content_type indicating value "application/yang-data+xml"
        from the TEST_SYSTEM entity
    } then {
        the IUT sends a r204
        to the TEST_SYSTEM entity
    }
}
```

TP Id	TP_mWT_PNC_PT_ETH_SVC_DELETE_BO_01
Test Objective	Verify that the IUT replies with an error code when the request is malformed.
Reference	ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause 5.3.2 "Exception Handling" ETSI GS mWT 024 [1], clause A.2.4 IETF RFC 6991 [i.11] "Common YANG Data Types" IETF RFC 8776 [i.12] "Common YANG Data Types for Traffic Engineering" draft-ietf-ccamp-client-signal-yang-05 [i.13] "A YANG Data Model for Transport Network Client Signals"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT
Initial Conditions	

with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a vRequestDelete containing
uri indicating value "/restconf/data/ietf-eth-tran-service:etht-svc/instances=ETHT_SVC_NAME" // Malformed URI
from the TEST_SYSTEM entity
} then {
the IUT sends a r400
to the TEST_SYSTEM entity
}
}	

TP Id	TP_mWT_PNC_ETH_PT_SVC_DELETE_BO_02
Test Objective	Verify that the IUT replies with an error code when it is requested to delete an unknown Bandwidth profile.
Reference	ETSI GS mWT 024 [1], clause 4.3.3 ETSI GS mWT 024 [1], clause 5.3.2 "Exception Handling" ETSI GS mWT 024 [1], clause A.2.4 IETF RFC 6991 [i.11] "Common YANG Data Types" IETF RFC 8776 [i.12] "Common YANG Data Types for Traffic Engineering" draft-ietf-ccamp-client-signal-yang-05 [i.13] "A YANG Data Model for Transport Network Client Signals"
Configuration	Config_mWT_2
PICS Selection	IUT_PNC and RESTCONF and SEC_TLS and ABSTRACT_PT
Initial Conditions	

with {
the IUT is idle and
the IUT having discovered several MW ethernet networks topologies
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a vRequestDelete containing
uri indicating value "/restconf/data/ietf-eth-tran-service:etht-svc/etht-svc-instances=UNKNOWN_ETHT_SVC_NAME"
from the TEST_SYSTEM entity
} then {
the IUT sends a r404
to the TEST_SYSTEM entity
}
}	

History

Document history		
V1.1.1	December 2024	Publication